

Yuta Oshima

PH.D. STUDENT · THE UNIVERSITY OF TOKYO

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Summary

I'm a first-year PhD student at The University of Tokyo, advised by Yutaka Matsuo, and closely collaborated with Hiroki Furuta.

My ultimate goal is to build **real-time, interactive world models** that seamlessly manifest human imagination into reality—enabling anyone to instantly craft and navigate their envisioned worlds with intuitive, precise control.

To move toward this vision, my research focuses on video diffusion models and world models. This includes recent work on **alignment** and **subject-driven generation** to enhance fine-grained controllability. Furthermore, I address the fundamental challenge of **memory mechanisms** to ensure long-horizon consistency in generative environments.

Education

The University of Tokyo

PH.D. IN COMPUTER SCIENCE AND ENGINEERING

Advisor: Prof. Yutaka Matsuo

Tokyo, Japan

Apr. 2025 - Present

The University of Tokyo

M.S. IN COMPUTER SCIENCE AND ENGINEERING

Tokyo, Japan

Apr. 2023 - Mar. 2025

The University of Tokyo

B.S. IN COMPUTER SCIENCE AND ENGINEERING

Tokyo, Japan

Apr. 2019 - Mar. 2023

Selected Publications

CONFERENCE

Inference-Time Text-to-Video Alignment with Diffusion Latent Beam Search

YUTA OSHIMA, MASAHIRO SUZUKI, YUTAKA MATSUO, HIROKI FURUTA

NeurIPS 2025

2025

ADOPT: Modified Adam Can Converge with Any β_2 with the Optimal Rate

SHOHEI TANIGUCHI, KENO HARADA, GOUKI MINEGISHI, YUTA OSHIMA, SEONG CHEOL JEONG, GO NAGAHARA, TOMOSHI IYAMA,

MASAHIRO SUZUKI, YUSUKE IWASAWA, YUTAKA MATSUO

NeurIPS 2024

2024

CONFERENCE WORKSHOP

AKITalk: Audio-Implicit Keypoints for Identity-Preserving Talking-Head Video Synthesis

RIKU TAKAHASHI, RONGZHI LI, YUTA OSHIMA, SHO KUNO, RYUGO MORITA, ISSEY SUKEDA

3rd Place in The ACM Multimedia Asia 2025 Grand Challenge

MMAAsia 2025 Workshop

2025

SSM Meets Video Diffusion Models: Efficient Video Generation with Structured State Spaces

YUTA OSHIMA, SHOHEI TANIGUCHI, MASAHIRO SUZUKI, YUTAKA MATSUO

ICLR 2024 Workshop

2024

Tactile In-Hand Pose Estimation through Perceptual Inference

TATSUYA KAMIJO, TOMOSHI IYAMA, YUTA OSHIMA, GENTIANE VENTURE, TATSUYA MATSUSHIMA, YUTAKA MATSUO, YUSUKE

IWASAWA

IROS 2023 Workshop

2023

Spotlight Presentation

PREPRINT

WorldPack: Compressed Memory Improves Spatial Consistency in Video World Modeling

YUTA OSHIMA, YUSUKE IWASAWA, MASAHIRO SUZUKI, YUTAKA MATSUO, HIROKI FURUTA

Preprint

Dec. 2025

Work Experience

EQUES Inc.
AI RESEARCHER
Research on anime-style video generation within the GENIAC project, supported by Japan's Ministry of Economy, Trade and Industry (METI).

Tokyo, Japan
Jul. 2025 - Present

World-leading Innovative Graduate Study Program Co-designing Future Society (WINGS-CFS), The University of Tokyo
RESEARCH ASSISTANT

Tokyo, Japan
Dec. 2023 - Present

Academic Honors

AWARDS

2025

3rd Place, The ACM Multimedia Asia 2025 Grand Challenge: Multimodal Multiethnic Talking-Head Video Generation

Kuala Lumpur, Malaysia

2024

Outstanding Presentation Award, JSAI Annual Conference

Shizuoka, Japan

2023

Spotlight Presentation, IROS 2023 Workshop on World Models and Predictive Coding in Cognitive Robotics

Detroit, U.S.A

GRANTS

2025

1,000,000 JPY, Leadership Development Program for Ph.D. (LDPP), the University of Tokyo

Academic Activities

2026

Reviewer, The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)

2026

Reviewer, International Conference on Learning Representations (ICLR)

DECEMBER 21, 2025

YUTA OSHIMA · RÉSUMÉ

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